Prompt

You are a Senior Linux Administrator and DevOps Engineer.

I have already completed SSH Passwordless Authentication between two Linux EC2 instances.

Teach me RSYNC in a concise, practical, and interview-focused manner.

Cover only:

1. What is RSYNC?
2. Why RSYNC exists
3. How RSYNC works
4. How RSYNC uses SSH
5. Difference between SCP and RSYNC

Then teach:

Server1 → Server2 file transfer using RSYNC

For each command provide:

- Command
- Command breakdown
- Practical example
- Verification

Cover:

- Single file transfer
- Multiple file transfer
- Directory transfer
- Updating changed files only
- Preserving permissions and ownership
- Resume interrupted transfer

Include important options:

- -a
- -v
- -z
- -P
- -e ssh

Finally provide:

- Advantages of RSYNC
- Limitations of RSYNC
- 10 interview questions with answers

Move step by step and wait for my confirmation before proceeding.
Focus on Linux administration and AWS EC2 examples only.
Do not include unnecessary theory or advanced topics.

RSYNC

🚀 STEP 1: RSYNC Fundamentals (Interview Core)

1. What is RSYNC?

rsync (Remote Sync) is a Linux utility used to:

- Efficiently **transfer and synchronize files/directories**
- Between:
  - Local ↔ Local
  - Local ↔ Remote (EC2)
  - Remote ↔ Remote (EC2 ↔ EC2)

👉 Key idea:
It does **NOT copy everything blindly** — it copies only differences.

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2. Why RSYNC exists

Before rsync, tools like `scp`:

- Always copied the **entire file again**
- Wasted bandwidth and time
- Not suitable for large files or frequent updates

🔥 Problem rsync solves:

- "Why re-send 5GB file if only 5MB changed?"

So rsync was built for:

- ⚡ Fast incremental transfers
- 💰 Low bandwidth usage (important in AWS EC2)
- 🔁 Resume capability
- 📂 Efficient backups & deployments

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3. How RSYNC works (Core Logic)

rsync uses a smart algorithm:

🧠 Working principle:

1. Compares source and destination files
2. Breaks files into **blocks**
3. Checks **checksum/delta differences**
4. Transfers only **changed blocks**

👉 So instead of copying whole file:

- It sends only **diff (delta)**

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4. How RSYNC uses SSH

In real DevOps / AWS EC2:

rsync usually runs over **SSH**:

🔐 Why SSH?

- Encryption (secure transfer)
- Authentication using key pair (passwordless setup you already did)
- Uses port **22**

🔗 Format:

rsync -e ssh source destination

👉 `-e ssh` = “use SSH as transport layer”

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5. SCP vs RSYNC (Interview Favorite ⚡)

Feature	SCP	RSYNC
Speed	Slower for large files	Faster (delta transfer)
Transfer type	Full copy every time	Incremental (only changes)
Resume support	❌ No	✅ Yes

Bandwidth usage	High	Low
Use case	Simple copy	Backup, sync, deployments
AWS usage	Basic transfers	Production deployments

🧠 SUMMARY (1-line interview answer)

👉 **rsync is a fast, secure file synchronization tool that transfers only modified data over SSH, making it more efficient than SCP.**

🚀 STEP 2: IMPLEMENTATION STEPS

Perfect 👍—this is exactly what makes rsync *real DevOps ready*. Let's do both cases cleanly with **EC2-style practical steps + verification**.

🖥️ PART A: Local ↔ Local (Same machine)

This is the **simplest rsync use-case** (no SSH involved).

🔹 1. Single File (Local → Local)

📌 Command

rsync -avz file1.txt /home/user/backup/

🧠 Breakdown

- `file1.txt` → source file
- `/home/user/backup/` → destination folder
- `-a` (archive mode)→ Preserve everything like a backup copy.
- `-v` (verbose mode) → Give logs of transfer.
- `-z` (compression)→ Compress data during transfer.

⚙️ Practical Example

echo "local test file" > file1.txt

mkdir backup

rsync -avz file1.txt backup/

🔍 Verification

ls backup/

cat backup/file1.txt

🔹 2. Directory (Local → Local)

📌 Command

rsync -avz mydir/ backup/mydir/

🧠 Key Point

- / at end means → **copy contents of directory**

⚙️ Example

mkdir mydir

touch mydir/a.txt mydir/b.txt

rsync -avz mydir/ backup/mydir/

🔍 Verify

ls -R backup/mydir

🔥 Use Case (Interview)

👉 Local rsync is used for:

- backups
- log rotation copies
- staging deployments

🖥️ PART B: Local ↔ Remote EC2 (IMPORTANT 🔥)

Now we use **SSH + EC2 instance**

🖥️ Assumptions

- Local machine = your laptop / EC2 workstation
- Remote = EC2 instance
- Key file: `mykey.pem`
- EC2 user: `ec2-user`
- EC2 IP: `13.x.x.x`

🔹 1. Local → EC2 (Upload to Server)

📌 Command

rsync -avz -e "ssh -i mykey.pem" file1.txt ec2-user@13.x.x.x:/home/ec2-user/

🧠 Breakdown

- `rsync` → sync tool
- `-a` → preserve metadata
- `-v` → verbose
- `-z` → compression
- `-e "ssh -i mykey.pem"` → SSH with private key
- `file1.txt` → local file
- `ec2-user@IP:/home/ec2-user/` → EC2 destination

⚙️ Practical Example

echo "hello ec2" > file1.txt

rsync -avz -e "ssh -i mykey.pem" file1.txt ec2-user@13.x.x.x:/home/ec2-user/

🔍 Verification (EC2)

ssh -i mykey.pem ec2-user@13.x.x.x

cat /home/ec2-user/file1.txt

🔹 2. EC2 → Local (Download from Server)

📌 Command

rsync -avz -e "ssh -i mykey.pem" ec2-user@13.x.x.x:/home/ec2-user/file2.txt .

🧠 Breakdown

- Source = EC2 file path
- Destination = `.` (current local directory)

⚙️ Practical Example (EC2 side first)

ssh -i mykey.pem ec2-user@13.x.x.x

echo "from server" > file2.txt

exit

Now on local:

rsync -avz -e "ssh -i mykey.pem" ec2-user@13.x.x.x:/home/ec2-user/file2.txt .

🔍 Verification (Local)

ls

cat file2.txt

🔥 KEY INTERVIEW INSIGHT

👉 In AWS DevOps:

Direction	Use Case
Local → EC2	Deploy code / upload builds
EC2 → Local	Logs / backups / artifacts

🚀 MINI SUMMARY

Local → Local

rsync -avz source destination

Local → EC2

rsync -avz -e "ssh -i key.pem" file ec2-user@IP:/path

EC2 → Local

rsync -avz -e "ssh -i key.pem" ec2-user@IP:/path/file .

🖥️ PART C: Server1 to Server2

Assume:

- 🖥️ Server1 (Source EC2): `ec2-user@server1`
- 🖥️ Server2 (Destination EC2): `ec2-user@server2`
- SSH passwordless already configured ✅

🔹 1. Single File Transfer

📌 Command

rsync -avz -e ssh file1.txt ec2-user@server2:/home/ec2-user/

🧠 Breakdown

- `rsync` → sync tool
- `-a` → archive mode (permissions, timestamps, symlinks preserved)
- `-v` → verbose output
- `-z` → compression during transfer
- `-e ssh` → use SSH tunnel
- `file1.txt` → source file

- `server2:/home/ec2-user/` → destination EC2 path

⚙️ Practical Example (Server1)

echo "Hello RSYNC" > file1.txt
rsync -avz -e ssh file1.txt ec2-user@server2:/home/ec2-user/

🔍 Verification (Server2)

ls -l /home/ec2-user/
cat file1.txt

🔹 2. Multiple File Transfer

📌 Command

rsync -avz -e ssh file1.txt file2.txt file3.txt ec2-user@server2:/home/ec2-user/

🧠 Breakdown

- Multiple files listed before destination
- Same flags apply
- Transfers all listed files in one SSH session (efficient)

⚙️ Practical Example

touch file1.txt file2.txt file3.txt
rsync -avz -e ssh file1.txt file2.txt file3.txt ec2-user@server2:/home/ec2-user/

🔍 Verification

ssh ec2-user@server2 "ls -l /home/ec2-user/"

🔹 3. Directory Transfer (VERY IMPORTANT 🔥)

📌 Command

rsync -avz -e ssh mydir/ ec2-user@server2:/home/ec2-user/mydir/

🧠 Breakdown

- `mydir/` → contents of directory
- Destination creates same structure
- `-a` ensures recursive copy

⚙️ Practical Example

mkdir mydir
touch mydir/a.txt mydir/b.txt

rsync -avz -e ssh mydir/ ec2-user@server2:/home/ec2-user/mydir/

🔍 Verification

ssh ec2-user@server2 "ls -R /home/ec2-user/mydir"

🔹 4. Updating Only Changed Files (RSYNC SUPER POWER ⚡)

📌 Command

rsync -avz -e ssh mydir/ ec2-user@server2:/home/ec2-user/mydir/

🧠 What happens?

- First run → full copy
- Second run → only changes transferred

⚙️ Example

echo "new change" >> mydir/a.txt

rsync -avz -e ssh mydir/ ec2-user@server2:/home/ec2-user/mydir/

🔍 Verification

ssh ec2-user@server2 "cat /home/ec2-user/mydir/a.txt"

🔹 5. Preserve Permissions & Ownership

📌 Command

rsync -avz -e ssh mydir/ ec2-user@server2:/home/ec2-user/mydir/

🧠 Key point:

- `-a` already includes:
  - permissions
  - timestamps
  - symbolic links
  - recursive copy

👉 That's why `-a` is critical in production.

🔹 6. Resume Interrupted Transfer 🔥

📌 Command

rsync -avzP -e ssh largefile.iso ec2-user@server2:/home/ec2-user/

🧠 Breakdown

- `-P` = combination of:
  - `--progress` (show progress)
  - `--partial` (resume incomplete files)

⚙️ Example scenario:

- Transfer 2GB file
- Connection breaks at 60%
- Re-run same command → resumes from 60%

🔍 Verification

ssh ec2-user@server2 "ls -lh /home/ec2-user/largefile.iso"

🔥 IMPORTANT OPTIONS SUMMARY

Option	Meaning
`-a`	archive mode (most important)
`-v`	verbose output
`-z`	compression
`-P`	progress + resume
`-e ssh`	use SSH

🧠 MINI INTERVIEW LINE

👉 “rsync with -a and -P over SSH is used in production to ensure secure, incremental, and resumable file transfers between EC2 instances.”

🚀 STEP 3: RSYNC — Production + Interview Master Layer

🔹 1. Advantages of RSYNC (Very Important for Interviews)

⚡ 1. Fast & Efficient

- Transfers only **changed data (delta sync)**
- Saves bandwidth (critical in AWS EC2)

🔐 2. Secure (with SSH)

- Works over SSH (`-e ssh`)
- Uses key-based authentication
- Encrypted transfer

🔁 3. Incremental Sync

- First run → full copy
- Next runs → only differences

👉 Perfect for:

- deployments
- backups
- config updates

📦 4. Resume Support

- Interrupted transfer resumes using `-P`
- No restart from zero

🧠 5. Preserves File Metadata

With `-a`:

- permissions
- timestamps
- ownership
- symlinks

💰 6. Cost Efficient in AWS

- Less network usage → lower transfer cost
- Faster EC2-to-EC2 operations

🔹 2. Limitations of RSYNC

❌ 1. Not ideal for real-time sync

- It is **not event-driven**
- Needs manual or cron trigger

❌ 2. Not for huge distributed systems

- Not designed for:
    - multi-node cluster sync
    - real-time replication systems

❌ 3. SSH dependency overhead

- SSH encryption adds slight overhead vs raw transfer

❌ 4. Not GUI friendly

- CLI-based only (not beginner friendly for some teams)

🔥 3. SCP vs RSYNC (Deep Interview Answer)

🆚 Core Difference:

Feature	SCP	RSYNC
Transfer type	Full copy	Incremental sync
Speed	Slower	Faster
Resume support	❌ No	✅ Yes
Bandwidth usage	High	Low
Best use case	One-time copy	Backup & deployment
Production usage	Rare	Very common

🧠 Interview One-Liner:

👉 “SCP is for simple secure copying, while rsync is for efficient, incremental synchronization used in production systems.”

🚀 4. 10 Interview Questions (VERY IMPORTANT)

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1. What is rsync?

👉 A tool used to efficiently sync files between local and remote systems by transferring only differences.

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2. How is rsync different from scp?

👉 rsync transfers only changed data; scp copies full files every time.

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3. Which protocol does rsync use?

👉 SSH for secure communication.

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4. What does `-a` option do?

👉 Archive mode: preserves permissions, timestamps, ownership, and copies recursively.

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5. How does rsync optimize transfer?

👉 By using delta-transfer algorithm (only changed blocks are sent).

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6. Can rsync resume failed transfers?

👉 Yes, using `-P` option.

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7. Is rsync suitable for large files?

👉 Yes, it is highly efficient for large files due to incremental transfer.

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8. What is the role of `-e ssh` in rsync?

👉 It tells rsync to use SSH as the transport layer.

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9. Where is rsync commonly used in AWS?

👉 EC2 deployments, backups, AMI data sync, and server migrations.

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10. Is rsync real-time sync tool?

👉 No, it is not real-time; it runs on-demand or scheduled.

🎯 FINAL SUMMARY (Interview Ready)

👉 **rsync = secure + fast + incremental + production-grade file sync tool used heavily in Linux & AWS environments for efficient data transfer.**
